

$$+\lambda[w_1 l_1 + \frac{1}{1+r} w_2 l_2 - c_1 - \frac{1}{1+r} c_2] \quad \dots (12.17)$$

In equation (12.17), the choice variables of the household are c_1, c_2, l_1 and l_2 . Our aim is to show the responsiveness of the relative labour supply due to the variation in relative wage in the two periods. Hence, we need to find out the first-order conditions for l_1 and l_2 from equation (12.17). These conditions are:

$$\frac{b}{1-l_1} = \lambda w_1 \quad \dots (12.18)$$

$$\frac{e^{-\rho} b}{1-l_2} = \frac{1}{1+r} \lambda w_2 \quad \dots (12.19)$$

In order to simplify the above two equations, we divide both sides of (12.18) by w_1 and both sides of (12.19) by $w_2/(1+r)$. Thus, we obtain

$$\frac{b}{1-l_1} \frac{1}{w_1} = \lambda \text{ and } \frac{e^{-\rho} b}{1-l_2} \frac{1+r}{w_2} = \lambda \quad \dots (12.20)$$

$$\text{Now, } \frac{e^{-\rho} b}{1-l_2} \frac{1+r}{w_2} = \frac{b}{1-l_1} \frac{1}{w_1}$$

The relative labour supply in the two periods is:

$$\frac{1-l_1}{1-l_2} = \frac{b}{e^{-\rho} b(1+r)} \frac{w_2}{w_1} = \frac{1}{e^{-\rho}(1+r)} \frac{w_2}{w_1} \quad \dots (12.21)$$

In the two-period case, the wage rate enters into the labour supply function. Equation (12.21) implies that there is a relation between the relative labour supply and the relative wage. This is to note here that $(1-l_1)$ means the household releases l_1 that means the remaining $(1-l_1)$ refers to the leisure in the first period. If w_1 rises relative to w_2 , then l_1 has to be increased or $(1-l_1)$ has to be decreased to satisfy equation (12.21). That means if the wage rate increases in the first period relative to the second period, labour supply has to increase in the first period relative to the second period (or leisure has to decrease in the first period relative to the second period). The elasticity of substitution between leisure in the two time periods is 1 (due to its logarithmic functional form).

Let us consider the case of a rise in r in equation (12.21). It indicates that there is an increase in the labour supply in the first period relative to second-period. A rise in interest rate in the first period raises the incentive to work more (hence, earn more and save more) in the first period relative to the second period. You might have observed that the effect of interest rate is crucial to fluctuation in employment in an economy. The effect of relative wage and interest rate on the labour supply (or employment or leisure) is known as intertemporal substitution in labour supply.

Check Your Progress 2

- 1) Give four examples of negative real shocks to an economy?

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- 2) Explain the concept of intertemporal substitution of labour.

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- 3) Explain how relative labour supply is dependent upon relative wage.

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- 4) In an intertemporal utility function, what is the impact of a rise in interest rate on labour supply?

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12.6 IMPACT OF SUPPLY SHOCKS TO THE ECONOMY

Now let us discuss the impact of the productivity shocks or technology shocks through our RBC model as we discussed in the previous section. Assume that there is an exogenous and temporary (lasting for one time period) favourable shock to technology in a given time period (t). This is reflected in our model by a rise in the A_t term as in equation (12.1). Let us denote the initial level technology as A_{0t} which increased to A_{1t} due to the shock. The growth in technology will cause a rise in the income Y_t at the given levels of K_t and L_t .

The effect of this shock is illustrated in Fig. 12.2. Initially, at the level of technology A_{0t} the corresponding production function is $K_t^\alpha (A_{0t} L_t)^{1-\alpha}$. Let us assume that L_0 is the optimal amount of labour at this level of technology and corresponding equilibrium output is Y_0 . The beneficial productivity shock (or technology shock) shifts the production function upward to $K_t^\alpha (A_{1t} L_t)^{1-\alpha}$. In addition to this upward shift, the nature of the shock is assumed to be such that the production function becomes steeper for any level of labour input. We know from microeconomics that the slope of the production function is the marginal product of labour. Thus, the positive technology shock increases marginal productivity also. Hence, at the same level of labour input (L_0), we have higher output Y'_1 due to the increase in productivity. That means the favourable shock has changed the production possibilities as well. If we increase the labour input to L_1 , the corresponding higher output is Y_1 due to the increased productivity.

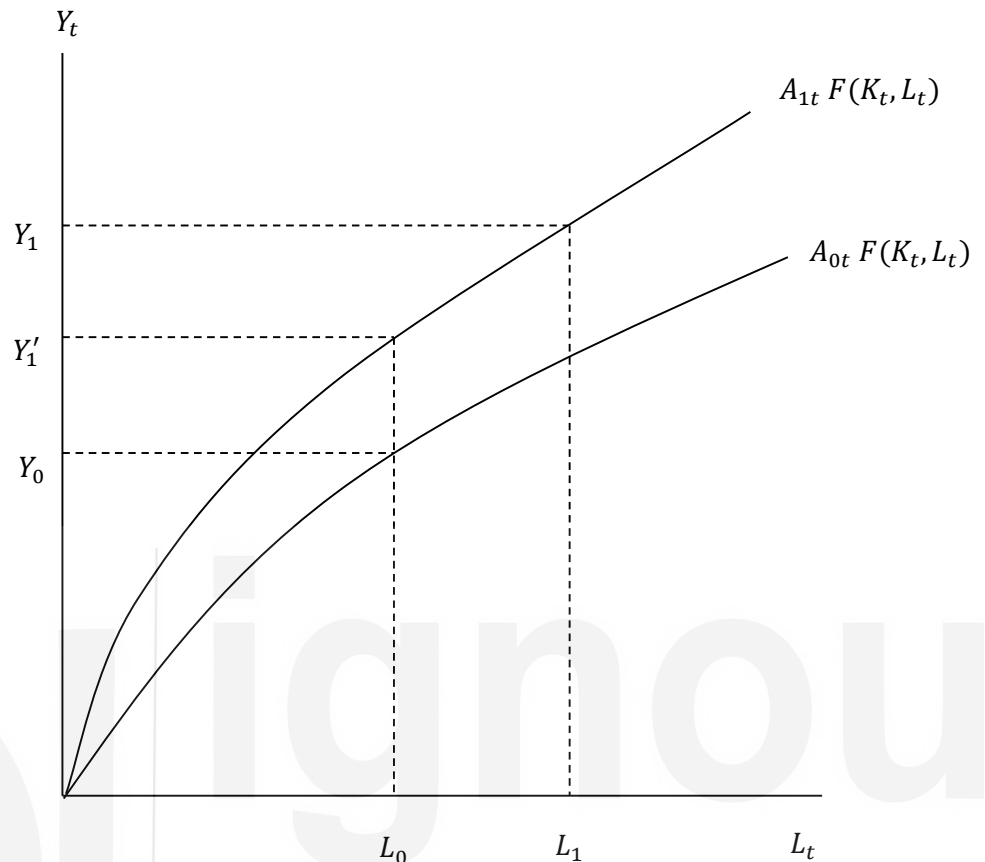


Fig. 12.2: Effect of Positive Technology Shock

Now, let us discuss about the distribution of such increased output. The increased output is distributed between consumption and saving. One possibility is that the increased output goes completely towards consumption. But, particularly in the case of a temporary shock, we may not prefer to consume all, but we like to save a portion of them for future higher consumption. If this is the case, equation (12.2) suggests that the higher the saving higher will be the level of investment. This will cause the capital stock to be higher in the next period as well as other future periods. Thus, the exogenous and temporary technology shock leads to increases in future capital stock.

Had the shock lasted for several periods or caused permanent effects, the economy would have responded somewhat differently. A permanent shock causes higher output for a longer period. Hence, the incentive to save would be reduced and consumption would be increased. The other possibility in the permanent shock case is that the increase in the labour input may not be high. Nevertheless, a persistent productivity shock will cause an increase in output, capital stock, and employment.

You should note that there is persistence of the effect of technology shocks. However, this statement is criticized by Keynesian economists as it cannot explain the persistence of real-world business cycles. The believers of RBC model argue that as explained in the model, the dynamic responses of optimizing

agents to change in economic conditions will have sustained effects. Such responses can explain periods of persistently high or low economic activity.

12.7 NEW-KEYNESIAN VIEW ON BUSINESS CYCLE

An important assumption of new-classical models is continuous market clearing, which implies perfect competition and flexibility in wages and prices. Thus, adjustments in prices and wages were instantaneous. Recall that classical economics was based on the same assumptions. Keynes discarded such assumptions and stressed on market imperfections. He advocated an activist role for the government.

A major criticism of Keynesian economics by Robert Lucas was that Keynesian economics does not have microfoundation. Further, expectations about future prices are not taken into account in Keynesian economics. New-Keynesian macroeconomics overcomes the above criticisms by including microfoundation and rational expectations.

What will happen if the above two features are brought into Keynesian economics? It implies that New-Keynesian economics also uses DSGE models. However, New-Keynesian models retain market imperfections and rigidities in prices and wages. Thus, there is not much difference between new-classical macroeconomics and new-Keynesian macroeconomics so far as methodology is concerned. The assumptions about market clearing are different however. New-Keynesian economists see certain imperfections in the market, particularly imperfect competition in the market and rigidities in prices and wages. In order to bring in such imperfections to DSGE models, new-Keynesian economists include various price-setting and wage-setting equations. These models take care of adjustment costs (such as menu costs) in wage setting, mark-up in price setting (as a measure of market imperfection) and some form of Taylor Rule for interest rate setting (policy rule).

Check Your Progress 3

- 1) Explain the impact of a supply shock on the economy through a diagram.

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- 2) State the issues emphasised by new-Keynesian economics in dynamic stochastic general equilibrium models.

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12.8 LET US SUM UP

Real business cycle (RBC) theory is based on microeconomic foundations in the sense that it takes into account the behaviour of economic agents such as firms and households. Thus, it considers the production function for analyzing changes in behavior of firms during a business cycle. Similarly, it considers the consumption function, particularly substitutability between labour and leisure on the part of households. Further, RBC theory is based on rational expectations in the sense that economic agents optimize on their objective functions by taking into account all information available to them. The RBC models adopt a general equilibrium framework for empirical estimation. The assumptions of RBC models, particularly its assumption of continuous market clearing, are challenged by new-Keynesian economics. The new-Keynesian economists, while adopting microfoundation, rational expectations and general equilibrium framework, incorporate market imperfections such as adjustment costs, staggered prices and mark-up pricing in dynamic stochastic general equilibrium (DGSE) models.

12.11 ANSWER/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) There are three basic assumptions in new-classical models, viz., rational expectations, continuous market clearing, and Lucas supply function. You should discuss the implications of these assumptions. Refer to Section 12.2 for details.
- 2) There are basically reasons, viz., inter-temporal substitution between labour and leisure, and Lucas supply function. According to new-classical economics, business cycles are rational response of economic agents to shocks to the economy. Refer to Section 12.2.

Check Your Progress 2

- 1) There could be several examples of real shocks. The negative shocks for example could be decline in technology (i.e., technical regress), unexpected rise in input prices (crude oil crisis), scanty rainfall (severe drought), and large-scale increase in taxes on inputs. Refer to Section 12.3.
- 2) Refer to Section 12.4 and explain on the basis of equation (12.21).
- 3) Refer to equation (12.21). You can find that the ratio of labour supply in both the periods is influenced by the ratio of wage rates.
- 4) Refer to Section 12.4 and explain on the basis of equation (12.21).

Check Your Progress 3

- 1) Refer to Fig. 12.2 and answer.
- 2) Refer to Section 12.7 and answer.